

THOHOYANDOU, South Africa

WMO: 681830

Lat: 23.0797S

Lon: 30.3836E

Elev: 614

StdP: 94.16

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	8.1	-4.2	2.9	21.6	-1.9	3.5	21.8	6.9	20.3	6.1	20.2	3.3	260	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	35.9	19.4	34.4	19.6	33.1	19.8	23.8	29.5	23.2	28.7	22.6	28.1	3.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	18.3	26.0	21.6	17.6	25.5	21.1	17.0	25.0	74.9	29.8	72.3	28.7	70.0	28.2	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.5	5.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	3.0	24.9	1.5	0.8	1.9	25.5	1.0	25.9	0.1	26.3	-1.0	26.9	(5)
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.3	24.5	24.7	23.6	20.9	19.0	16.8	16.3	18.5	21.1	22.5	23.8	24.2
	DBStd	4.00	2.43	2.20	2.25	2.29	2.22	2.35	2.35	2.68	3.06	3.32	3.47	3.57
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	207	1	0	1	6	17	57	72	30	10	6	4	4
	CDD10.0	4131	450	413	423	326	280	204	195	265	332	389	413	441
	CDD18.3	1296	192	179	165	82	39	11	8	37	92	136	168	187
	CDH23.3	11967	1517	1408	1212	589	423	207	199	537	1107	1402	1645	1722
CDH26.7	4754	557	527	437	179	106	37	41	178	499	658	757	780	
Wind	WSAvg	2.5	2.5	2.3	2.2	2.2	2.4	2.7	2.5	2.6	2.7	2.8	2.6	2.4
Precipitation	PrecAvg	1066	224	204	146	51	20	16	19	14	26	56	114	159
	PrecMax	2591	671	1217	468	174	106	67	106	50	126	163	275	333
	PrecMin	600	31	29	27	0	0	0	0	0	0	4	28	71
	PrecStd	432	144	236	109	40	23	18	23	12	32	43	53	79
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	35.8	35.2	32.4	32.1	29.6	29.8	32.7	36.6	37.8	37.1	36.8
		MCWB	20.8	21.0	20.9	19.1	17.3	15.1	13.5	16.4	16.7	18.1	19.1	21.5
	2%	DB	33.0	33.1	32.7	30.7	29.4	27.6	27.7	30.7	34.0	35.2	35.1	34.8
		MCWB	21.5	21.7	21.1	19.5	16.9	14.5	13.4	15.5	16.4	18.3	19.3	21.0
	5%	DB	31.5	31.6	31.1	28.8	27.8	25.9	25.6	28.8	31.9	33.1	33.2	33.0
		MCWB	21.2	21.5	21.2	19.4	16.6	14.4	13.1	14.7	16.2	18.2	19.8	21.1
	10%	DB	29.9	30.1	29.3	27.0	25.9	23.9	23.3	26.5	29.8	30.5	31.2	31.1
		MCWB	21.4	21.4	21.2	19.0	16.4	14.2	13.3	14.5	16.1	18.2	19.9	21.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	25.4	24.1	22.3	20.8	19.0	16.4	18.2	19.6	21.6	23.2	24.1
		MCDB	30.9	29.8	29.6	28.2	26.9	24.1	21.8	27.2	28.4	30.7	29.2	30.2
	2%	WB	23.6	23.9	23.2	21.3	19.3	17.1	15.5	17.2	18.7	20.7	22.6	23.2
		MCDB	29.6	28.5	28.5	26.5	25.2	22.3	21.5	25.6	27.3	29.4	28.7	29.3
	5%	WB	22.8	23.0	22.5	20.7	18.3	16.1	14.8	16.5	18.0	20.0	21.8	22.5
		MCDB	28.0	28.3	27.8	25.6	24.1	21.5	20.8	24.4	26.7	28.0	28.4	28.8
	10%	WB	22.2	22.2	21.9	20.0	17.4	15.2	14.1	15.7	17.2	19.2	21.1	21.8
		MCDB	27.1	27.3	27.0	24.9	23.1	20.7	20.5	23.0	25.4	26.7	27.1	28.4
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.4	9.6	10.6	13.2	14.0	13.2	13.9	13.9	12.1	11.0	10.0
		MCDBR	12.4	12.4	13.5	14.9	16.4	17.1	17.6	18.4	18.9	17.4	15.4	14.2
		MCWBR	3.9	3.9	4.6	5.9	6.4	7.1	6.7	6.5	6.4	5.4	4.4	4.1
	5% WB	MCDBR	9.9	9.6	10.3	10.9	12.9	13.0	12.8	14.4	13.8	12.9	11.6	11.2
	MCWBR	3.8	4.0	4.3	4.8	6.2	6.5	6.2	5.7	5.6	4.6	4.4	3.8	
Clear-Sky Solar Irradiance	taub	0.401	0.396	0.386	0.369	0.335	0.322	0.325	0.376	0.444	0.453	0.419	0.412	
	taud	2.390	2.403	2.424	2.426	2.466	2.481	2.467	2.310	2.154	2.179	2.311	2.356	
	Ebn at Noon	944	935	917	889	881	874	884	868	848	872	921	935	
	Edh at Noon	129	125	117	108	95	89	94	119	151	155	139	134	
All-Sky Solar Radiation	RadAvg	6.14	5.97	5.36	4.63	4.36	3.87	4.06	4.81	5.43	5.74	5.96	6.04	
	RadStd	0.48	0.59	0.46	0.32	0.21	0.28	0.19	0.26	0.38	0.39	0.52	0.59	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page